

MIXTURE TO BEAMFORMED MIXTURE: LEVERAGING BEAMFORMED MIXTURE AS WEAK-SUPERVISION FOR SPEECH ENHANCEMENT AND NOISE-ROBUST ASR

Zhong-Qiu Wang and Ruizhe Pang

Department of Computer Science and Engineering
Southern University of Science and Technology, Shenzhen, China

{wang.zhongqiu41, ruizhepang}@gmail.com

ABSTRACT

In multi-channel speech enhancement and robust automatic speech recognition (ASR), beamforming can typically improve the signal-to-noise ratio (SNR) of the target speaker and produce reliable enhancement with little distortion to target speech. With this observation, we propose to leverage beamformed mixture, which has a higher SNR of the target speaker than the input mixture, as a weak supervision to train deep neural networks (DNNs) to enhance the input mixture. This way, we can train enhancement models using pairs of real-recorded mixture and its beamformed mixture, and potentially realize better generalization to real mixtures, compared with only training the models on simulated mixtures, which usually mismatch real mixtures. Evaluation results on the real-recorded CHiME-4 dataset show the effectiveness of the proposed algorithm.

Index Terms— Weakly-supervised speech enhancement.

1. INTRODUCTION

Deep learning has dramatically advanced speech enhancement [1–4]. The most widely-adopted approach to date is supervised deep learning, where DNNs are trained on simulated pairs of clean speech and its mixture with interference signals (such as noises, reverberation, and concurrent speech) to predict the clean speech based on the simulated mixture [1–4]. Although strong performance is observed on simulated mixtures, many studies report that, even with large-scale training [1, 5–7], where massive mixtures are simulated to cover many variations during training, this approach has limited generalizability to real-recorded mixtures [8–12]. This is largely because the current simulation techniques cannot simulate mixtures that are sufficiently realistic. As a result, simulated training mixtures are often mismatched with real test mixtures, consequently resulting in limited generalizability.

One potential solution is leveraging real mixtures for model training. However, real mixtures are typically unlabeled and labeling them at the sample level is nearly impossible. Supervised learning based approaches hence cannot be applied. In this context, unsupervised methods [13–18] have been proposed. However, due to not leveraging any supervision, their performance and application scenarios are often limited.

Various types of weak supervision such as the ones afforded by source prior models [19, 20], adversarially-trained discriminators [21], target-sound classifiers [22] and ASR models [23] have

been designed for training models directly on mixtures. However, many types of weak supervision are not at the sample level and do not offer fine-grained supervision. In multi-channel speech enhancement, beamforming can usually produce reliable enhancement to target speech and boost the SNR of the target speaker. Intuitively, beamformed mixtures could be utilized as a *sample-level* weak supervision to train DNNs to enhance the observed, lower-SNR input mixtures.

In this context, we propose *mixture to beamformed mixture* (M2BM) training. It builds upon the recent *mixture to mixture* (M2M) algorithm [24], where DNNs are trained to produce an estimate for target speech and another for non-target signals such that the two estimates, after being linearly filtered and summated, can approximate observed mixtures. M2BM extends M2M by further encouraging that the two estimates can be linearly filtered and summated to approximate beamformed mixtures. This way, we could leverage the weak-supervision in beamformed mixtures for enhancing the lower-SNR input mixtures. Following SuperM2M [25], which combines M2M training on real mixtures with supervised learning on simulated mixtures, we combine M2BM training with supervised training, leading to SuperM2BM. Evaluation results on the popular CHiME-4 dataset show the effectiveness of SuperM2BM.

It should be noted that there are studies [26–29] leveraging conventional signal processing algorithms such as spatial clustering and beamforming to derive pseudo-labels for training separation or enhancement models directly on mixtures. However, the DNN estimates are trained to directly approximate the pseudo-labels, which are typically not sufficiently accurate. Therefore, their performance is limited by that of conventional algorithms. Differently, in M2BM, the two DNN estimates are trained to respectively approximate the target and non-target components in beamformed mixtures, and hence could yield better performance.

2. PHYSICAL MODEL AND OBJECTIVES

In a noisy-reverberant environment with a P -microphone far-field microphone array and a single target speaker (illustrated in the bottom-right plot of Fig. 1), the physical model of the recorded far-field mixture can be formulated, in the short-time Fourier transform (STFT) domain, as follows:

$$Y_p(t, f) = X_p(t, f) + V_p(t, f), \quad (1)$$

where $p \in \{1, \dots, P\}$ indexes the P far-field microphones, t indexes T frames, f indexes F frequency bins, and $Y_p(t, f)$, $X_p(t, f)$, and $V_p(t, f)$ are respectively the mixture, reverberant speech of the

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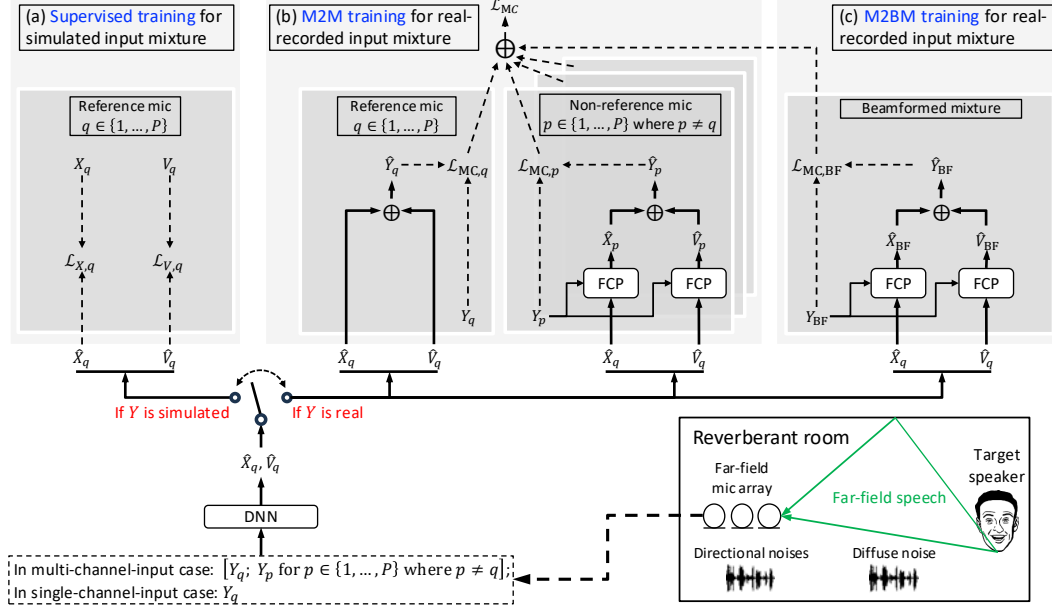


Fig. 1: Illustration of SuperM2BM, which trains the same DNN-based enhancement model based on (a) supervised training if the input mixture is simulated; and (b) M2M training and (c) M2BM training if the input mixture is real-recorded.

target speaker, and reverberant signals of the other sources captured by microphone p at time t and frequency f . When dropping indices p , t and f , we refer to the corresponding spectrograms. We designate microphone $q \in \{1, \dots, P\}$ as the reference microphone. Assuming weak reverberation, we aim at suppressing environmental noises and reconstruct target reverberant speech at the reference microphone (i.e., X_q) based on the multi-channel input mixture Y .

3. REVIEW OF SUPERM2M

SuperM2BM builds on SuperM2M [25], which combines supervised training on simulated mixtures and M2M training on real mixtures to train the same DNN for enhancement. This section reviews SuperM2M.

3.1. Review of Supervised Training

Fig. 1(a) illustrates supervised training on simulated mixtures. The DNN is trained to predict the target and non-target signals at the reference microphone q . The estimates are denoted as $\hat{X}_q$ and $\hat{V}_q$. Since the mixture is simulated, we can define a supervised loss for each estimate:

$$\mathcal{L}_{X,q} = \frac{1}{\sum_{t,f} |Y_q(t,f)|} \sum_{t,f} \mathcal{G}(X_q(t,f), \hat{X}_q(t,f)), \quad (2)$$

$$\mathcal{L}_{V,q} = \frac{1}{\sum_{t,f} |Y_q(t,f)|} \sum_{t,f} \mathcal{G}(V_q(t,f), \hat{V}_q(t,f)), \quad (3)$$

where, let $|\cdot|$ extracts magnitude and $\mathcal{R}(\cdot)$ and $\mathcal{I}(\cdot)$ respectively extract real and imaginary components, $\mathcal{G}(\cdot, \cdot)$ is defined as

$$\mathcal{G}(a, b) = |\mathcal{R}(a) - \mathcal{R}(b)| + |\mathcal{I}(a) - \mathcal{I}(b)| + ||a| - |b||. \quad (4)$$

3.2. Review of M2M Training

Fig. 1(b) illustrates M2M training on real mixtures. Since real mixtures are unlabeled, we cannot apply supervised loss to penalize the

DNN estimates $\hat{X}_q$ and $\hat{V}_q$. In M2M training [24, 25], the two estimates are constrained such that they can be linearly filtered to reconstruct the mixture captured at each microphone, by optimizing the so-called mixture-constraint (MC) loss, which can be defined as

$$\mathcal{L}_{MC} = \mathcal{L}_{MC,q} + \frac{1}{P-1} \times \sum_{p=1, p \neq q}^P \mathcal{L}_{MC,p}, \quad (5)$$

where $\mathcal{L}_{MC,q}$ and $\mathcal{L}_{MC,p}$ are respectively the MC losses at the reference microphone q and non-reference microphone p . $\mathcal{L}_{MC,q}$ constrains that the two estimates should add up to the mixture at the reference microphone q (i.e., Y_q):

$$\begin{aligned} \mathcal{L}_{MC,q} &= \sum_{t,f} \mathcal{F}(Y_q(t,f), \hat{Y}_q(t,f)) \\ &= \sum_{t,f} \mathcal{F}(Y_q(t,f), \hat{X}_q(t,f) + \hat{V}_q(t,f)), \end{aligned} \quad (6)$$

where $\mathcal{F}(Y_q(t,f), \hat{Y}_q(t,f)) = \frac{\mathcal{G}(Y_q(t,f), \hat{Y}_q(t,f))}{\sum_{t',f'} |Y_q(t',f')|}$, with $\mathcal{G}(\cdot, \cdot)$ defined in (4). $\mathcal{L}_{MC,p}$ constrains that the two estimates, after being linearly filtered, should add up to the mixture at each non-reference microphone p (i.e., Y_p):

$$\begin{aligned} \mathcal{L}_{MC,p} &= \sum_{t,f} \mathcal{F}(Y_p(t,f), \hat{Y}_p(t,f)) \\ &= \sum_{t,f} \mathcal{F}(Y_p(t,f), \hat{X}_p(t,f) + \hat{V}_p(t,f)) \\ &= \sum_{t,f} \mathcal{F}(Y_p(t,f), \hat{\mathbf{h}}_p(f)^H \hat{\mathbf{X}}_q(t,f) + \hat{\mathbf{r}}_p(f)^H \hat{\mathbf{V}}_q(t,f)). \end{aligned} \quad (7)$$

In (7), $\hat{X}_p(t,f) = \hat{\mathbf{h}}_p(f)^H \hat{\mathbf{X}}_q(t,f)$, where $(\cdot)^H$ computes Hermitian transpose, $\hat{\mathbf{X}}_q(t,f) = [\hat{X}_q(t-I+1, f), \dots, \hat{X}_q(t+J, f)]^T \in \mathbb{C}^{I+J}$ stacks a window of $I+J$ T-F units, and $\hat{\mathbf{h}}_p(f) \in \mathbb{C}^{I+J}$ is the linear filter. Similarly, $\hat{V}_p(t,f) = \hat{\mathbf{r}}_p(f)^H \hat{\mathbf{V}}_q(t,f)$, where

$\widehat{\mathbf{V}}_q(t, f) = [\hat{V}_q(t - I + 1, f), \dots, \hat{V}_q(t + J, f)]^T \in \mathbb{C}^{I+J}$ and $\hat{\mathbf{r}}_p(f) \in \mathbb{C}^{I+J}$. The filter taps I and J are hyperparameters to tune.

To compute $\mathcal{L}_{\text{MC},p}$, the linear filters $\hat{\mathbf{h}}_p(f)$ and $\hat{\mathbf{r}}_p(f)$ need to be computed first. Both of them can be computed by using the forward convolutive prediction (FCP) algorithm [30]. For example, $\hat{\mathbf{h}}_p(f)$ can be computed as follows:

$$\hat{\mathbf{h}}_p(f) = \underset{\mathbf{h}_p(f)}{\operatorname{argmin}} \sum_t \frac{|Y_p(t, f) - \mathbf{h}_p(f)^H \widehat{\mathbf{X}}_q(t, f)|^2}{\hat{\lambda}_p(t, f)}. \quad (8)$$

$\hat{\lambda}$ is a weighting term defined as $\hat{\lambda}_p(t, f) = \xi \times \max(|Y_p|^2) + |Y_p(t, f)|^2$, where ξ (tuned to 10^{-2}) floors the weighting term and $\max(\cdot)$ extracts the maximum value of a power spectrogram. The same method can be used to compute the other filter $\hat{\mathbf{r}}_p(f)$. Notice that the optimization problem in (8) is quadratic, and a closed-form solution can be readily computed and plugged into (7) to compute the loss and train the DNN.

3.3. Review of SuperM2M Training

SuperM2M [25] combines supervised and M2M training, forming a semi-supervised approach that can train the same DNN-based enhancement model based on both simulated and real mixtures. If the input mixture is simulated, supervised training is used, while if it is real, M2M training is used. This way, the model can readily learn signal patterns from massive simulated mixtures, and, meanwhile, see real mixtures during training so that it can generalize better to real mixtures.

In SuperM2M [25], for real mixtures, the number of microphones used in input can be smaller than that used in loss computation. For example, we can train a DNN using monaural input while computing the loss on all the microphones. At run time, the DNN performs monaural enhancement.

4. SUPERM2BM

We propose SuperM2BM, which improves SuperM2M in Fig. 1(a) and (b) by including a new loss defined on beamformed mixtures. This section describes M2BM training, and how to obtain beamformed mixtures for M2BM training.

4.1. Mixture to Beamformed Mixture (M2BM)

Fig. 1(c) illustrates M2BM training on real mixtures. For each real training mixture, let us assume that we can first compute a beamforming result Y_{BF} . We will describe how to compute it later in Section 4.2. Since, when the number of microphones is sufficiently large, Y_{BF} would have a much higher SNR of the target speaker than the input mixture recorded by any of the microphones, it could be leveraged as a weak supervision to train the enhancement model. In detail, we consider the beamforming result as a mixture signal recorded by a *virtual microphone*, and compute an additional MC loss, which follows $\mathcal{L}_{\text{MC},p}$ in (7), for model training:

$$\begin{aligned} \mathcal{L}_{\text{MC,BF}} &= \sum_{t,f} \mathcal{F}(Y_{\text{BF}}(t, f), \hat{Y}_{\text{BF}}(t, f)) \\ &= \sum_{t,f} \mathcal{F}(Y_{\text{BF}}(t, f), \hat{X}_{\text{BF}}(t, f) + \hat{V}_{\text{BF}}(t, f)) \\ &= \sum_{t,f} \mathcal{F}(Y_{\text{BF}}(t, f), \hat{\mathbf{h}}_{\text{BF}}(f)^H \widehat{\mathbf{X}}_q(t, f) + \hat{\mathbf{r}}_{\text{BF}}(f)^H \widehat{\mathbf{V}}_q(t, f)), \end{aligned} \quad (9)$$

where $\widehat{\mathbf{X}}_q(t, f)$ and $\widehat{\mathbf{V}}_q(t, f)$ are defined in the same way as those in (7), and the linear filters are computed in the same way as (8). Building on the loss in (5), the training loss for real mixtures becomes

$$\mathcal{L}_{\text{MC}} = \mathcal{L}_{\text{MC},q} + \frac{1}{P-1} \times \sum_{p=1, p \neq q}^P \mathcal{L}_{\text{MC},p} + \mathcal{L}_{\text{MC,BF}}. \quad (10)$$

4.2. Deriving Beamformed Mixtures for M2BM Training

So far, we assume that a beamforming result can be first computed from real-recorded mixtures and later used for M2BM training. We employ the method proposed in [31] to compute the beamformer. We first train a monaural supervised speech enhancement model based on simulated mixtures to predict the target and non-target signals, following the method shown in Fig. 1(a). Next, we apply the model to enhance the real mixture captured by each microphone in the training set, obtaining $\hat{\mathbf{X}}(t, f) = [\hat{X}_1(t, f), \dots, \hat{X}_P(t, f)]^T$ and $\hat{\mathbf{V}}(t, f) = [\hat{V}_1(t, f), \dots, \hat{V}_P(t, f)]^T$. Assuming that sound sources are non-moving, we compute time-invariant spatial covariance matrices for the target and non-target signals via

$$\hat{\Phi}_{\text{X}}(f) = \sum_t \hat{\mathbf{X}}(t, f) \hat{\mathbf{X}}(t, f)^H, \quad (11)$$

$$\hat{\Phi}_{\text{V}}(f) = \sum_t \hat{\mathbf{V}}(t, f) \hat{\mathbf{V}}(t, f)^H, \quad (12)$$

where $(\cdot)^H$ computes Hermitian transpose. Next, we compute the relative transfer function $\hat{\mathbf{c}}_q(f)$ as follows:

$$\mathbf{r}(f) = \mathcal{P}(\hat{\Phi}_{\text{X}}(f)), \quad (13)$$

$$\hat{\mathbf{c}}_q(f) = \mathbf{r}(f)/r_q(f), \quad (14)$$

where $\mathcal{P}(\cdot)$ extracts the principal eigenvector. A time-invariant minimum variance distortionless response (MVDR) beamformer [2] is then computed as

$$\hat{\mathbf{w}}_q(f) = \frac{\hat{\Phi}_{\text{V}}(f)^{-1} \hat{\mathbf{c}}_q(f)}{\hat{\mathbf{c}}_q(f)^H \hat{\Phi}_{\text{V}}(f)^{-1} \hat{\mathbf{c}}_q(f)}, \quad (15)$$

and the beamforming result is obtained by

$$Y_{\text{BF}}(t, f) = \hat{\mathbf{w}}_q(f)^H \mathbf{Y}(t, f). \quad (16)$$

After obtaining beamformed mixtures for all the real training mixtures, we leverage them to train the SuperM2BM system, by including the $\mathcal{L}_{\text{MC,BF}}$ loss defined in (9).

Notice that the beamformed mixtures are computed by using a DNN trained on simulated mixtures in a supervised way. The trained DNN itself often exhibits limited generalizability to real mixtures, as the immediate outputs of the DNN (i.e., $\hat{\mathbf{X}}$ and $\hat{\mathbf{V}}$) are often not accurate on real mixtures [8–12]. Nonetheless, many existing studies [1, 2, 32–34] have shown that linear beamforming results derived based on the DNN estimates usually exhibit little distortion to target speech and can often result in much better ASR performance. This is largely because the beamforming result is obtained via linear filtering, which can introduce little distortion to target speech. Although time-invariant beamforming typically cannot suppress non-target signals aggressively, it can reliably improve the SNR of the target speaker. We could hence utilize the beamformed mixtures for M2BM training.

Like SuperM2M, SuperM2BM, once trained, does not need to compute beamformed mixtures for inference. It just runs feed-forwarding once and uses $\hat{X}_q$ as the enhancement result.

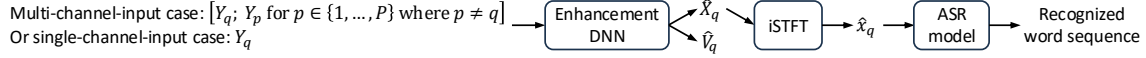


Fig. 2: Robust ASR pipeline.

5. EXPERIMENTAL SETUP

We evaluate SuperM2BM based on CHiME-4 [9, 35, 36], the most popular dataset to date for robust ASR and speech enhancement. It contains both simulated (SIMU) and real-recorded (REAL) mixtures for training and evaluation. The real mixtures are recorded by using a tablet mounted with 6 microphones (with the second one on the rear and the other 5 facing front) in 4 representative environments including buses, streets, pedestrian areas and cafeteria, where room reverberation and strong non-stationary, multi-source noises can naturally co-exist. During data collection, the target speaker hand-holds the tablet and utters text prompts shown on the tablet. In the training, validation and test sets, there are respectively 7, 138, 1, 640 and 1, 320 simulated mixtures, and 1, 600, 1, 640 and 1, 320 real mixtures. In CHiME-4, the reverberation is weak and the challenge is mainly in noise suppression. The sampling rate is 16 kHz.

Fig. 2 shows the evaluation pipeline for robust ASR, where enhanced speech is fed to a pre-trained backend ASR model for recognition. We employ the strongest ASR model [37] reported to date on CHiME-4. It is an encoder-decoder transformer model trained on WavLM features, using a transformer language model for decoding.

For enhancement, the STFT window size is 32 ms, hop size 8 ms, and the square root of the Hann window is used as the analysis window. TF-GridNet [38], which has shown strong performance in recent speech separation benchmarks, is employed as the DNN architecture. Using the symbols defined in Table I of [38], we set its hyperparameters to $D = 128$, $B = 4$, $I = 1$, $J = 1$, $H = 200$, $L = 4$ and $E = 4$. It is trained via complex spectral mapping [31, 39] to predict the real and imaginary (RI) components of the target and non-target signals based on the RI components of the input mixtures. The filter taps I and J used to define $\tilde{X}_q(t, f)$ and $\tilde{V}_q(t, f)$ in (7) and (9) are respectively tuned to 20 and 1. For each real training mixture, we use all the 6 microphones to compute the M2M loss, while the input to the DNN can be configured to 1-, 2- or 6-channel. The beamformed mixture is computed based on the front five microphones, considering that the rear microphone has a much lower SNR [36]. Word error rates (WER) and DNSMOS OVRL [40] are used as the evaluation metrics.

6. EVALUATION RESULTS

Let us first look at the ASR results in Table 1. Comparing row 0 with 2a, 4a and 6a, we observe that purely-supervised learning based systems (i.e., training via Fig. 1(a)) have severe generalization issues to the real test data (especially in the 2- and 6-channel input cases), although they performed reasonably well on the simulated test data. Compared with purely-supervised systems, SuperM2M, trained by combining Fig. 1(a) and (b), clearly improves the ASR performance on the real data, confirming the effectiveness of including real mixtures for model training. By including beamformed mixtures for loss computation, SuperM2BM, which combines the training methods in Fig. 1(a), (b) and (c), further improves the ASR performance. This confirms the effectiveness of M2BM. In addition, SuperM2BM outperforms IRIS [37] and MultiIRIS [41], the previous best.

Row 7 reports the ASR results of the beamformed test mixtures, which can indicate the quality of the beamformed mixtures used for M2BM training. The monaural supervised model in row 2a is

Table 1: Robust ASR and DNSMOS OVRL Results on CHiME-4.

Row	System	Input #mics	DNSMOS OVRL $\uparrow$		WER (%) $\downarrow$			
			Val.	Test	Val.		Test	
			REAL	REAL	SIMU	REAL	SIMU	REAL
0	Mixture [37]	1	1.52	1.39	5.93	4.03	8.25	4.47
1	IRIS [37]	1	-	-	3.16	2.03	6.12	3.92
2a	Supervised	1	3.33	3.18	3.53	2.15	8.05	4.83
2b	SuperM2M	1	3.30	3.14	3.29	2.05	6.92	3.60
2c	SuperM2BM	1	3.25	3.14	3.39	1.82	6.80	3.10
3	MultiIRIS [41]	2	-	-	2.04	1.66	2.04	2.65
4a	Supervised	2	2.99	2.83	1.54	11.93	2.29	23.18
4b	SuperM2M	2	1.81	1.64	1.57	2.71	2.22	2.85
4c	SuperM2BM	2	2.73	2.51	1.46	1.40	2.18	1.93
5	MultiIRIS [41]	6	-	-	1.22	1.33	1.24	1.77
6a	Supervised	6	2.38	2.11	0.83	41.76	1.31	67.26
6b	SuperM2M	6	1.84	1.63	0.83	2.42	1.34	2.26
6c	SuperM2BM	6	2.48	2.17	0.85	1.25	1.34	1.56
7	1ch supervised +5ch beamform.	5	2.48	2.19	0.93	1.22	1.39	1.55

used to derive the beamformed mixtures, using the method described in Section 4.2. Comparing 6c and 7, we observe that 6-channel SuperM2BM obtains ASR results comparable to the beamforming method. From 2b and 2c, and 4b and 4c, we observe clear ASR improvement by using the beamformed mixtures for training.

Table 1 also reports DNSMOS scores. From 2b and 2c, 4b and 4c, and 6b and 6c, we observe that SuperM2BM obtains comparable or better DNSMOS scores than SuperM2M. This confirms the effectiveness of using beamformed mixtures for training. Comparing 2c, 4c and 6c with 7, we observe that SuperM2BM obtains clearly better or comparable DNSMOS scores. This suggests that the enhancement performance of SuperM2BM is not limited by that of beamformed mixtures. Although 6-channel SuperM2BM obtains WER comparable to 5-channel beamforming (i.e., row 6c and 7), the beamformed signals can help us train better 1-channel and 2-channel SuperM2BM models.

Notice that the SuperM2M results reported in Table 1 are worse than the ones reported in the original SuperM2M paper [25]. This is because, in [25], close-talk mixtures are assumed available for training and are leveraged to derive an additional loss to improve the training of SuperM2M models. In contrast, this study assumes that close-talk mixtures are not available for training. This assumption is more realistic, as collecting paired close-talk and far-field mixtures is much more difficult than collecting far-field mixtures alone.

7. CONCLUSIONS

We have proposed M2BM, which leverages beamformed mixture as weak supervision to train speech enhancement models on unlabeled real mixtures. We have combined M2BM with supervised learning on simulated mixtures, yielding SuperM2BM. Evaluation results on the CHiME-4 dataset show that SuperM2BM can effectively exploit the weak supervision in beamformed mixtures to improve generalizability. In closing, we highlight that, besides simulated mixtures, SuperM2BM only requires a set of multi-channel unlabeled real mixtures for training. In practical application development, one could just record some real mixtures from the target domain and leverage SuperM2BM to realize better generalizability.

8. REFERENCES

- [1] D. Wang and J. Chen, "Supervised Speech Separation Based on Deep Learning: An Overview," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 26, no. 10, pp. 1702–1726, 2018.
- [2] R. Haeb-Umbach, T. Nakatani, M. Delcroix, C. Boeddeker, and T. Ochiai, "Microphone Array Signal Processing and Deep Learning for Speech Enhancement: Combining Model-Based and Data-Driven Approaches to Parameter Estimation and Filtering," *IEEE Signal Processing Magazine*, vol. 41, no. 6, pp. 12–23, 2025.
- [3] S. Araki, N. Ito, R. Haeb-Umbach, G. Wichern, Z.-Q. Wang, and Y. Mitsufuji, "30+ Years of Source Separation Research: Achievements and Future Challenges," in *ICASSP*, 2025.
- [4] Y. Xie and Z.-H. Tan, "A Survey of Deep Learning for Complex Speech Spectrograms," *arXiv preprint arXiv:2505.08694*, 2025.
- [5] J. Chen, Y. Wang, S. E. Yoho, D. Wang, and E. W. Healy, "Large-Scale Training to Increase Speech Intelligibility for Hearing-Impaired Listeners in Novel Noises," *Journal of The Acoustical Society of America*, vol. 139, no. 5, pp. 2604–2612, 2016.
- [6] W. Zhang, K. Saijo, Z.-Q. Wang, S. Watanabe *et al.*, "Toward Universal Speech Enhancement for Diverse Input Conditions," in *ASRU*, 2023.
- [7] W. Zhang, J.-w. Jung, S. Watanabe, and Y. Qian, "Improving Design of Input Condition Invariant Speech Enhancement," in *ICASSP*, 2024.
- [8] T. J. Cox, J. Barker, W. Bailey, S. Graetzer, M. A. Akeroyd, J. F. Culling, and G. Naylor, "Overview of The 2023 ICASSP SP Clarity Challenge: Speech Enhancement For Hearing Aids," in *ICASSP*, 2023.
- [9] E. Vincent *et al.*, "An Analysis of Environment, Microphone and Data Simulation Mismatches in Robust Speech Recognition," *Computer Speech & Language*, vol. 46, pp. 535–557, 2017.
- [10] S. Watanabe, M. Mandel, J. Barker, E. Vincent *et al.*, "ChiME-6 Challenge: Tackling Multispeaker Speech Recognition for Unsegmented Recordings," in *arXiv preprint arXiv:2004.09249*, 2020.
- [11] F. Yu *et al.*, "M2MeT: The ICASSP 2022 Multi-Channel Multi-Party Meeting Transcription Challenge," in *ICASSP*, 2022, pp. 6167–6171.
- [12] Z. Wang, S. Wu, H. Chen *et al.*, "The Multimodal Information Based Speech Processing (MISP) 2022 Challenge: Audio-Visual Diarization And Recognition," in *ICASSP*, 2023, pp. 1–5.
- [13] S. Wisdom, E. Tzinis, H. Erdogan, R. J. Weiss, K. Wilson, and J. R. Hershey, "Unsupervised Sound Separation using Mixture Invariant Training," in *NeurIPS*, 2020.
- [14] E. Tzinis, Y. Adi, V. K. Ithapu, B. Xu, P. Smaragdis, and A. Kumar, "RemixIT: Continual Self-Training of Speech Enhancement Models via Bootstrapped Remixing," *IEEE J. Sel. Top. Signal Process.*, vol. 16, no. 6, pp. 1329–1341, 2022.
- [15] Y. Bando *et al.*, "Neural Full-Rank Spatial Covariance Analysis for Blind Source Separation," *IEEE Signal Process. Lett.*, vol. 28, pp. 1670–1674, 2021.
- [16] R. Aralikatti *et al.*, "Reverberation as Supervision for Speech Separation," in *ICASSP*, 2023.
- [17] Z.-Q. Wang and S. Watanabe, "UNSSOR: Unsupervised Neural Speech Separation by Leveraging Over-determined Training Mixtures," in *NeurIPS*, vol. 36, 2023, pp. 34 021–34 042.
- [18] Z.-Q. Wang, "USDnet: Unsupervised Speech Dereverberation via Neural Forward Filtering," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 32, pp. 3882–3895, 2024.
- [19] N. Zhang, J. Yan, and Y. Zhou, "Weakly Supervised Audio Source Separation via Spectrum Energy Preserved Wasserstein Learning," in *IJCAI*, 2018, pp. 4574–4580.
- [20] Z. Xu, X. Fan, Z.-Q. Wang, X. Jiang, and R. R. Choudhury, "Unsupervised Blind Speech Separation with a Diffusion Prior," in *ICML*, 2025.
- [21] D. Stoller, S. Ewert, and S. Dixon, "Adversarial Semi-Supervised Audio Source Separation Applied to Singing Voice Extraction," in *ICASSP*, 2018, pp. 2391–2395.
- [22] F. Pishdadian, G. Wichern, and J. Le Roux, "Finding Strength in Weakness: Learning to Separate Sounds with Weak Supervision," in *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 28, 2020, pp. 2386–2399.
- [23] X. Chang, W. Zhang, Y. Qian *et al.*, "MIMO-SPEECH: End-to-End Multi-Channel Multi-Speaker Speech Recognition," in *ASRU*, 2019, pp. 237–244.
- [24] Z.-Q. Wang, "Mixture to Mixture: Leveraging Close-Talk Mixtures as Weak-Supervision for Speech Separation," *IEEE Signal Process. Lett.*, vol. 31, pp. 1715–1719, 2024.
- [25] —, "SuperM2M: Supervised and Mixture-to-Mixture Co-Learning for Speech Enhancement and Robust ASR," *Neural Networks*, vol. 188, no. 107408, pp. 1–16, 2025.
- [26] L. Drude, D. Hasenklever, and R. Haeb-Umbach, "Unsupervised Training of A Deep Clustering Model for Multichannel Blind Source Separation," in *ICASSP*, 2019, pp. 695–699.
- [27] P. Seetharaman, G. Wichern, J. Le Roux, and B. Pardo, "Bootstrapping Single-Channel Source Separation via Unsupervised Spatial Clustering on Stereo Mixtures," in *ICASSP*, 2019, pp. 356–360.
- [28] E. Tzinis, S. Venkataramani, and P. Smaragdis, "Unsupervised Deep Clustering for Source Separation: Direct Learning from Mixtures Using Spatial Information," in *ICASSP*, 2019, pp. 81–85.
- [29] Y. Zhang, H. Zhang, and X. Zhang, "Single-channel Speech Enhancement Student under Multi-channel Speech Enhancement Teacher," in *APSIPA-ASC*, 2022, pp. 372–377.
- [30] Z.-Q. Wang *et al.*, "Convolutional Prediction for Monaural Speech Dereverberation and Noisy-Reverberant Speaker Separation," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 29, pp. 3476–3490, 2021.
- [31] —, "Complex Spectral Mapping for Single- and Multi-Channel Speech Enhancement and Robust ASR," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 28, pp. 1778–1787, 2020.
- [32] T. Yoshioka, N. Ito, M. Delcroix *et al.*, "The NTT ChiME-3 System: Advances in Speech Enhancement and Recognition for Mobile Multi-Microphone Devices," in *ASRU*, 2015, pp. 436–443.
- [33] J. Heymann, L. Drude, A. Chinaev, and R. Haeb-Umbach, "BLSTM Supported GEV Beamformer Front-End for The 3rd ChiME Challenge," in *ASRU*, 2015, pp. 444–451.
- [34] X. Zhang, Z.-Q. Wang, and D. Wang, "A Speech Enhancement Algorithm by Iterating Single- and Multi-Microphone Processing and Its Application to Robust ASR," in *ICASSP*, 2017, pp. 276–280.
- [35] J. Barker, R. Marxer, E. Vincent, and S. Watanabe, "The Third 'ChiME' Speech Separation and Recognition Challenge: Dataset, Task and Baselines," in *ASRU*, 2015, pp. 504–511.
- [36] —, "The Third 'ChiME' Speech Separation and Recognition Challenge: Analysis and Outcomes," *Computer Speech & Language*, vol. 46, pp. 605–626, 2017.
- [37] X. Chang, T. Maekaku *et al.*, "End-to-End Integration of Speech Recognition, Speech Enhancement, and Self-Supervised Learning Representation," in *Interspeech*, 2022, pp. 3819–3823.
- [38] Z.-Q. Wang, S. Cornell, S. Choi, Y. Lee, B.-Y. Kim, and S. Watanabe, "TF-GridNet: Integrating Full- and Sub-Band Modeling for Speech Separation," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 31, pp. 3221–3236, 2023.
- [39] K. Tan and D. Wang, "Learning Complex Spectral Mapping With Gated Convolutional Recurrent Networks for Monaural Speech Enhancement," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 28, pp. 380–390, 2020.
- [40] C. K. A. Reddy, V. Gopal, and R. Cutler, "DNSMOS: A Non-Intrusive Perceptual Objective Speech Quality Metric to Evaluate Noise Suppressors," in *ICASSP*, 2021, pp. 6493–6497.
- [41] Y. Masuyama, X. Chang, S. Cornell *et al.*, "End-to-End Integration of Speech Recognition, Dereverberation, Beamforming, and Self-Supervised Learning Representation," in *SLT*, 2023, pp. 260–265.